

Science Study Assistant – Training PDF

A compact science knowledge base designed for PDF retrieval, question answering, explanation generation, and quiz generation.

1. Scientific Method

- Science studies the natural world using observation, measurement, evidence, and logical reasoning.
- Typical steps are: observation → question → hypothesis → experiment or investigation → data collection → analysis → conclusion.
- A hypothesis is a testable explanation or prediction. A scientific conclusion should be based on evidence rather than personal belief.
- Variables in an experiment can include an independent variable, a dependent variable, and controlled variables.

2. Matter and Its Properties

- Matter is anything that has mass and occupies space. Common states of matter are solid, liquid, gas, and plasma.
- Solids have a definite shape and volume. Liquids have a definite volume but take the shape of their container. Gases have neither a definite shape nor a definite volume.
- Physical properties include mass, volume, density, color, melting point, and boiling point. A physical change does not create a new substance.
- Chemical properties describe how a substance can undergo chemical change. A chemical reaction produces one or more new substances.

3. Atoms and Elements

- An atom contains a nucleus surrounded by electrons. The nucleus contains protons and neutrons.
- Protons have positive electric charge, electrons have negative charge, and neutrons have no net electric charge.
- The atomic number of an element equals its number of protons.
- Elements are pure substances made of one type of atom. The periodic table organizes elements according to their atomic numbers and recurring properties.

4. Forces and Motion

- Force is a push or pull that can change an object's motion. The SI unit of force is the newton (N).
- Newton's second law is $F = ma$, where F is net force, m is mass, and a is acceleration.
- Speed describes how quickly distance is covered: $\text{speed} = \text{distance} / \text{time}$.
- Velocity includes both speed and direction. Acceleration is the rate at which velocity changes.

5. Work, Energy and Power

- Work is done when a force causes displacement in the direction of the force. For a constant force in the same direction, $W = Fd$.
- Kinetic energy is energy associated with motion: $KE = \frac{1}{2}mv^2$.
- Gravitational potential energy near Earth's surface can be represented as $PE = mgh$.
- Power is the rate of doing work: $P = W/t$. The SI unit of power is the watt (W).

6. Heat and Temperature

- Temperature is related to the average kinetic energy of particles, while heat is energy transferred because of a temperature difference.
- Heat can be transferred by conduction, convection, and radiation.
- Conduction is common in solids, convection involves bulk movement of fluids, and radiation transfers energy through electromagnetic waves.

7. Electricity

- Electric current is the rate of flow of electric charge. Its SI unit is the ampere (A).
- Voltage is electric potential difference and is measured in volts (V). Resistance is measured in ohms (Ω).
- Ohm's law for an ohmic conductor is $V = IR$.
- Electrical power can be calculated using $P = VI$.

8. Light and Sound

- Light is electromagnetic radiation and can travel through a vacuum.
- Reflection is the bouncing back of light from a surface. Refraction is the change in direction of light when it enters a different medium.
- Sound is a mechanical wave and generally requires a material medium to propagate.
- Frequency is the number of complete cycles per second and is measured in hertz (Hz).

9. Cells and Life Science

- The cell is the basic structural and functional unit of living organisms.
- Plant cells commonly contain a cell wall, cell membrane, cytoplasm, nucleus, chloroplasts, and a large central vacuole.
- Animal cells have a cell membrane, cytoplasm, nucleus, and other organelles but do not have a cell wall or chloroplasts.
- Photosynthesis converts light energy into chemical energy. A simplified representation is: carbon dioxide + water $\rightarrow$ glucose + oxygen, in the presence of light and chlorophyll.

10. Human Body

- The circulatory system transports substances such as oxygen, nutrients, hormones, and wastes through the body.

- The respiratory system is responsible for gas exchange. Oxygen enters the body and carbon dioxide is removed.
- The digestive system breaks food into smaller substances that can be absorbed and used by the body.
- The nervous system coordinates information processing and responses using electrical and chemical signals.

11. Ecology

- An ecosystem includes living organisms and their physical environment.
- Producers, such as green plants, make organic food using energy from sunlight. Consumers obtain energy by eating other organisms.
- Decomposers break down dead organic matter and return nutrients to the environment.
- Energy generally flows through food chains and food webs, while matter such as carbon and nitrogen cycles through ecosystems.

12. Earth and Space Science

- Earth has a layered structure commonly described as crust, mantle, outer core, and inner core.
- The water cycle includes processes such as evaporation, condensation, precipitation, infiltration, and runoff.
- The Solar System contains the Sun and objects that orbit it, including planets, dwarf planets, moons, asteroids, and comets.
- Earth's rotation produces the cycle of day and night, while its revolution around the Sun contributes to the yearly cycle.

13. Quick Reference Formulas

- Speed = distance / time
- Density = mass / volume
- $F = ma$
- $W = Fd$
- $KE = \frac{1}{2} mv^2$
- $PE = mgh$
- $P = W/t$
- $V = IR$
- Electrical power: $P = VI$

14. Practice Questions

Question	Answer
What is the atomic number of an element?	The number of protons in its atoms.
Which state of matter has a definite volume but takes the shape of its container?	Liquid.
What is the SI unit of force?	Newton (N).
Write Newton's second law.	$F = ma$.
What is the formula for speed?	speed = distance / time.
What is kinetic energy?	Energy associated with motion.
What are the three main modes of heat transfer?	Conduction, convection, and radiation.
State Ohm's law.	$V = IR$.
What is the basic structural and functional unit of life?	The cell.
What role do decomposers play in ecosystems?	They break down dead organic matter and help return nutrients to the environment.

15. Retrieval / Q&A; Notes

This document is intentionally organized into clear sections and short factual chunks. A study assistant can retrieve the most relevant section for a question and use it as context. For questions not supported by this document, the assistant should say that the information is not available in the supplied study material instead of inventing an answer.